SOLUTION

Question 1: Water Jug Problem

Solution Steps

1. Fill the **4-gallon jug**.
2. Pour water into the **3-gallon jug** until it is full.
   - Remaining in the 4-gallon jug = **1 gallon**.
3. Empty the **3-gallon jug**.
4. Pour the **1 gallon** from the 4-gallon jug into the 3-gallon jug.
5. Fill the **4-gallon jug** again.
6. Pour water into the **3-gallon jug** until it becomes full.
   - The 3-gallon jug already contains **1 gallon**, so it needs **2 gallons** more.
   - Therefore, **2 gallons remain in the 4-gallon jug**.

Final Answer: 2 gallons are obtained in the 4-gallon jug.

Question 2: Mars Rover Intelligent Agent

PEAS

- **Performance Measure:** Safe navigation, successful sample collection, efficient battery usage.
- **Environment:** Mars surface (rocks, sand, hills, craters).
- **Actuators:** Wheels, robotic arm, drill, communication system.
- **Sensors:** Cameras, proximity sensors, temperature sensors, battery sensors.

Operating Environment

- Partially Observable
- Deterministic
- Sequential
- Dynamic
- Continuous
- Single Agent

Actions

- Move forward/backward
- Turn left/right
- Avoid obstacles
- Collect samples
- Send data to Earth
- Recharge using solar panels

Performance Evaluation

- Distance covered
- Samples collected
- Battery efficiency
- Obstacle avoidance
- Successful data transmission

Agent Model

Learning Agent

Reason: It learns from experience and adapts to the changing Martian environment.

Question 3: 8-Queens Problem

Initial State

An empty 8 × 8 chessboard.

State Space

All possible arrangements of eight queens on the board.

Operators

Place one queen in a safe position in the next column.

Goal Test

No two queens attack each other in any row, column, or diagonal.

Path Cost

Each queen placement has a cost of 1; total cost is not important compared to finding a valid arrangement.

Search Strategy

Backtracking Search

Model: State Space Search Model

Question 4: OLA Cab Booking Intelligent Agent

Agent Model

Goal-Based Agent

Pseudocode

Start
Input source and destination
Display available cab types
Calculate fare and distance
Select the best available cab
Confirm booking
Display driver details
Stop

Reason

The system selects the best cab to achieve the customer's goal of reaching the destination efficiently.

Question 5: Uniform Cost Search (UCS)

Algorithm

1. Start from node **S**.
2. Insert **S** into the priority queue with cost 0.
3. Remove the node with the lowest cumulative cost.
4. Expand its neighboring nodes and update their cumulative costs.
5. Repeat until the goal node **G** is reached.
6. Return the least-cost path.

Node Expansion

- S (0)
- A (1)
- C (2)
- B (4)
- G (4)

Optimal Path

S → A → C → G

Total Cost

1 + 1 + 2 = 4

Model

State Space Search Model

Algorithm Used

Uniform Cost Search (UCS)

Result: All five analytical problems were solved successfully using the appropriate Artificial Intelligence models and search algorithms.